Macintosh PiForma
Owner's Guide



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Welcome to the Macintosh PiForma

This little computer began in the basement of Apple headquarters, which is remarkable, because Apple headquarters did not have a basement. A rogue engineer came across an Apple IIc Monitor down there, emptied it, and Frankensteined a computer into the case from whatever parts were lying around, several of which had not, strictly speaking, been invented yet.

The work went on under the nose of the vice president of the day, a man with firm opinions about customers opening their computers. In short, they shouldn't. The engineer was in no hurry to show him a Macintosh with easy access to its insides, running open-source software and other people's good ideas.

Then, in 1985, the vice president was let go, and the mood changed. The CEO turned out to be rather more open-minded about unconventional ideas than anyone had expected, and saw in the PiForma an experiment in something nobody had a name for yet: a Macintosh clone, built from someone else's parts. Clones would not officially arrive for another ten years; this one simply got there first. The project came up out of the basement, and the PiForma was given the green light.

Only one PiForma was ever made, and this is it. Its story is almost entirely unknown; you may be the first person ever to read it.

Inside, the old monitor's case holds a modern display and a battery-powered computer, with a familiar-looking desktop, games, and a few diversions that are there mostly because they're fun. Its name is a small joke; its carrying bag is a surprisingly serious undertaking. You can use it without knowing how the wiring works, although you will want to know which way the keyboard goes back into the bag.

This guide explains what the PiForma does, what is inside it and how it works. It starts with the things you need to touch, connect, and switch on. Later chapters explain the parts and programs behind the appearance. There is no required course in Raspberry Pi administration before your first game.

About this manual

Chapter 1, "Getting Started," explains how to unpack the computer, arrange the equipment, start up, and stop, and what to do if it won't start. It also puts everything back into the bag in the right order. Taking things out is easier than remembering where they belonged.

Chapter 2, "Inside the Macintosh PiForma," introduces the monitor shell, Raspberry Pi, display, battery, controls, keyboard, mouse, speakers, and bag. It explains what the parts do before getting into the finer details.

Chapter 3, "How the Macintosh PiForma Works," introduces the desktop, launcher, menu bar, games, and other programs. It distinguishes the computer's physical controls from the menus on its screen.

Chapter 4, "Caring for Your Macintosh PiForma," covers handling, cleaning, charging, storage, and transport. It includes the brief but comprehensive description of the service department.

Appendix A collects technical information and points to the project documentation. **Appendix B** helps you find a useful check when something goes wrong. The **glossary** explains unfamiliar terms used in the guide.

Start with Chapter 1 for your first session. Return to Chapter 3 when you want to work with programs and files, or to Appendix B when something goes wrong.

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CHAPTER 1

Getting Started

This chapter will help you unpack your Macintosh PiForma, set up the keyboard, mouse and other accessories, and start the computer. You will also learn how to shut down properly, return everything to the bag, and keep the battery charged. The packing order matters more than you might expect.



Unpacking the equipment

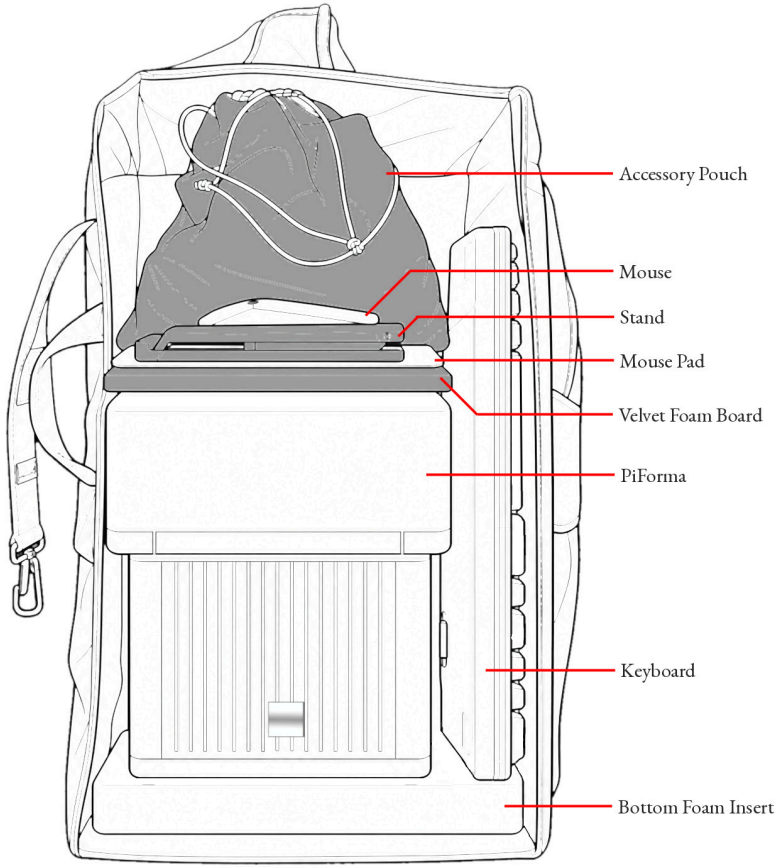
Set the carrying bag upright on a stable surface. Open the clasp, take it out of the strap loop, and pull the loop back out through the grommets; "Closing the bag," near the end of this chapter, shows the clasp. Leave room nearby for each item as you take it out.

1. **Remove the small brown accessory pouch near the top.** It contains two game controllers, the USB hub, and the power cable. Set the pouch and its contents aside.
2. **Take out the mouse and the collapsible computer stand beneath the pouch.**
3. **Lift out the mouse pad beneath them.**
4. **Remove the velvet-covered foam board beneath the mouse pad.** Keep it flat; it protects the computer when everything is packed.

The PiForma is underneath the board, with its screen facing upward. The keyboard is tucked against one side of the computer. It is not another layer in the stack. Note which way it faces, as described under "The keyboard's place" below, and lift the keyboard out first, before the computer.

Lift the PiForma by its handle and case. Do not push on the screen: pressing on the display can damage the screen or its mounts.

Unpacking the equipment



Arrangement in the carrying bag.

Finding the handle

The carrying handle may be hidden against the inside front wall of the bag. Slide a hand down inside that wall, along the top side of the PiForma, until you can pull the handle outward. Do not pull on a connector or an edge of the display to lift the machine.

If lifting it out upright is awkward, lay the bag on its back first and try the gentler removal angle. Lift carefully. There is no prize for wrestling it out upright.

The keyboard's place

Notice the keyboard's orientation before moving it: the keys face the bag wall, the thicker edge faces the rear of the bag, and the thinner edge faces the front. Remember these three directions when you pack it again.

The molded edges in the bottom foam locate the back of the computer. The first photograph shows the empty base; the second shows the PiForma and keyboard in place. Match the keyboard's orientation, rather than merely finding a gap large enough to hold it.



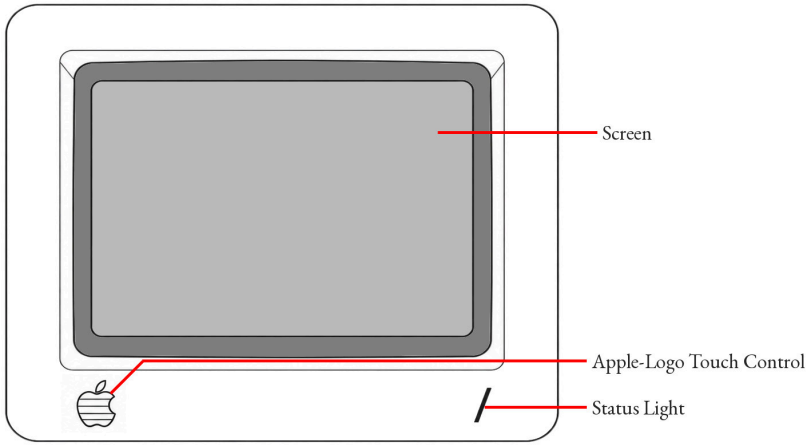
The bottom foam insert. Seat the back of the PiForma against the molded edges.



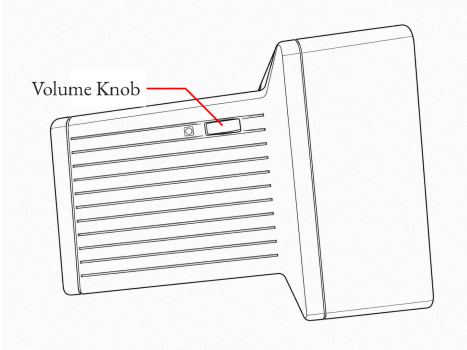
PiForma and keyboard inside the bag. Keys face the bag wall; the thick edge goes toward the rear, the thin edge toward the front.

Taking a closer look

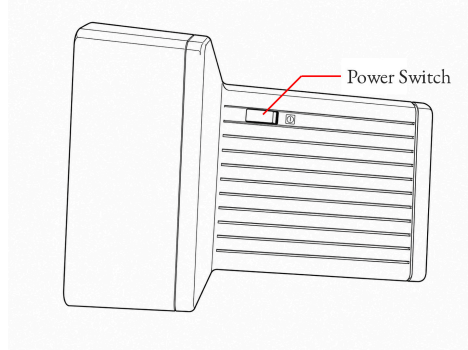
Before connecting anything, find the power switch on the right side, the green status light at the front, and the chime-selector dial at the back. The Apple logo at the front is also a touch control. It is separate from the physical power switch.



■ The front: screen, Apple-logo touch control, and status light.



■ The side volume knob.

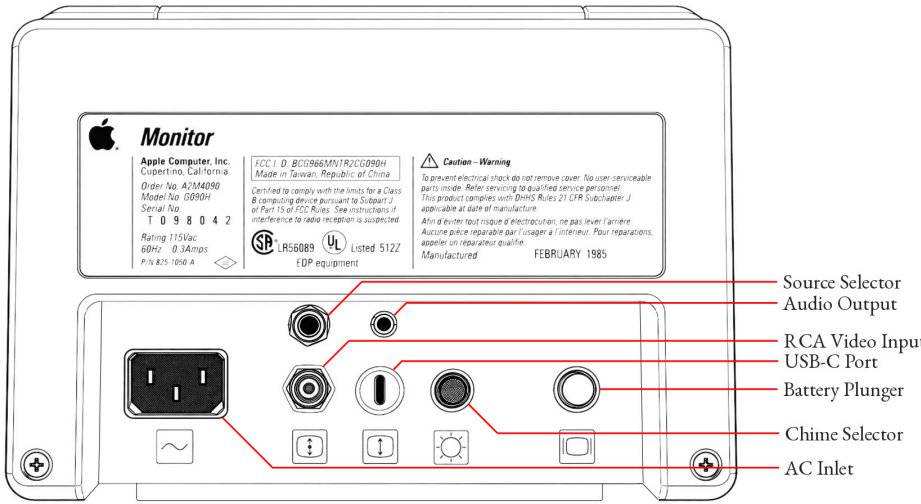


■ The power switch on the opposite side. This is separate from the Apple-logo touch control.

The one USB port at the back is a USB-C port. A single device plugged straight into it needs a USB-C plug of its own or an adapter. For more than one device, or for older USB plugs, use the USB hub.

Source selector: The rear button that moves the screen to its next input: HDMI for the Raspberry Pi, AV1 for the RCA input.

At the rear are the source selector, a composite-video RCA input and an audio output socket. The RCA jack carries video only; the audio socket sends sound out to external speakers or headphones. Press the source selector until the screen shows HDMI for the Raspberry Pi picture. Chapter 2 explains the other inputs.

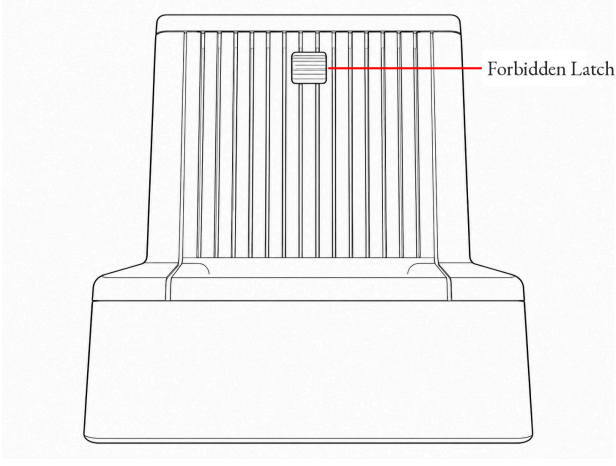


■ Rear controls and connections. The RCA input is composite video; the audio socket is an output for speakers or headphones.

The forbidden latch

The slider on top opens the upper compartment. Around here it is known as the Forbidden Latch. Management has specifically instructed us not to tell you how to open it. They would prefer that you admire it without touching it. You can check the battery's charge through the top slits without opening anything, which keeps management happy.

Anyway, instructions for opening it are on page 23, in the section called "A closer look, strictly off the record," toward the end of this chapter.



The top of the PiForma and its Forbidden Latch.

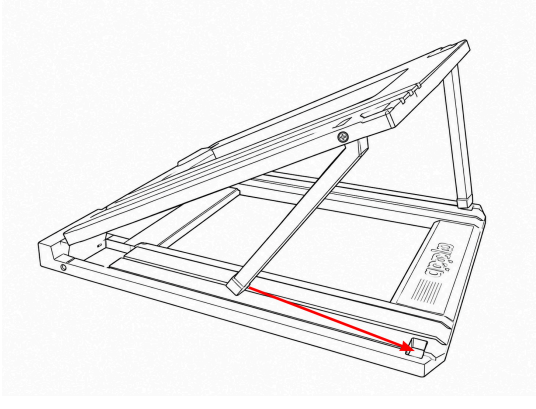
Basic ergonomics

Choose a solid, level table with room for the stand, keyboard, mouse pad, and cables. Place the screen at a comfortable viewing angle, out of glare, with the keyboard directly in front of you and the mouse beside it. Put speakers where they will not tip, block the screen, or pull on the computer's cables. Adjust your chair rather than spending the afternoon leaning toward the computer.

Putting it all together

The stand, keyboard and mouse

1. **Open the collapsible stand and place it on the level table.** Seat the support leg in its final position, as the arrow in the illustration shows. Check that it is stable before adding the computer.
2. **Lower the PiForma onto the stand, exactly in the center.** Check that it rests securely rather than leaning toward one side.
3. **Put the keyboard directly in front.** Turn it on with the switch on its right-hand edge.
4. **Put the mouse on its pad beside the keyboard.** Its power switch is underneath; turn it on if necessary.



Seat the support leg in the position indicated by the arrow, then check the stand before adding the computer.

Center the PiForma exactly: not to the left, not to the right, but in the middle of the stand as you set it down. Seen from the side, it should rest on the stand as in the second photograph.



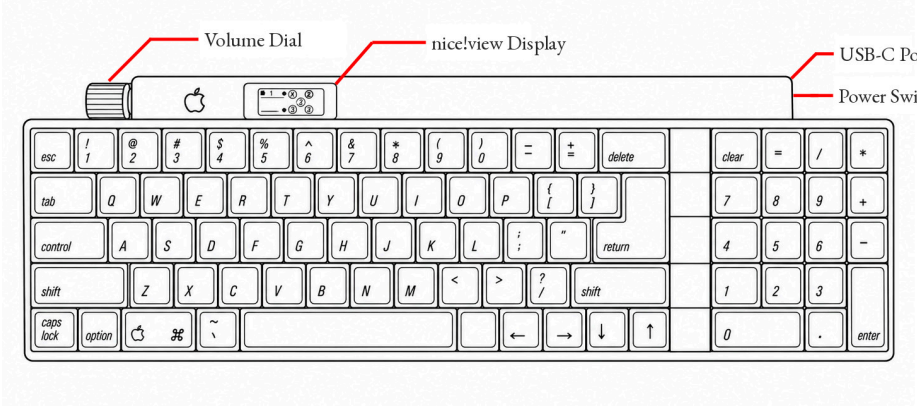
Lower the PiForma straight down onto the middle of the stand.



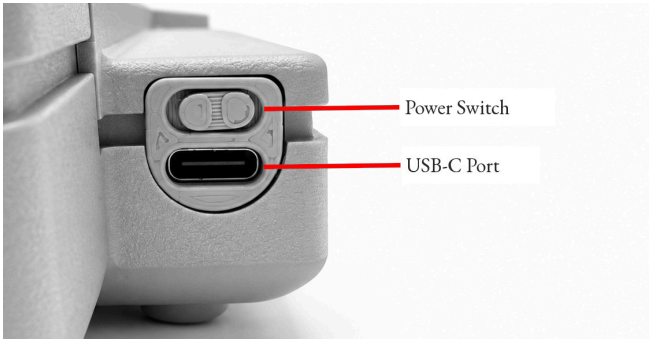
Seen from the side: the PiForma resting on its stand.

The keyboard controls

The overhead view identifies the keyboard's volume dial, on the left, and its nice!view status display. The power switch and USB-C charging port are on the right-hand edge, where the keyboard's ADB port used to be. Their labels in the overhead view show only the general location, because the port and switch themselves are not visible from above; the close-up of the right edge shows them. Charge the keyboard from that USB-C port.



Keyboard controls. The USB-C port and power-switch callouts indicate their general location, not a directly visible socket or switch.



The keyboard's right-hand edge: power switch and USB-C charging port.

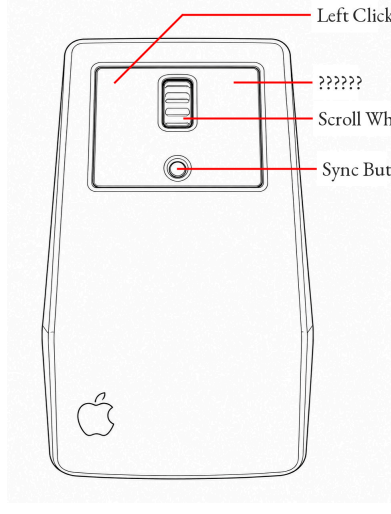
The volume dial on the keyboard does the same job as the knob on the side of the PiForma. There are two so that you can change the volume during a console game with the keyboard packed away, or reach the nearer one when the keyboard is in front of you. The nice!view display shows the keyboard's battery level and connection; Chapter 2 explains it. You don't need to press it.

Caps Lock deserves a warning. It is a vintage locking key, and on the PiForma it behaves like a Shift key held down. While it is down, letters come out as capitals and the number keys type symbols, and the keyboard keeps sending Shift, which runs down its battery. Make sure Caps Lock is up, level with Shift and Option, before you start and before you pack the keyboard away.

The mouse controls

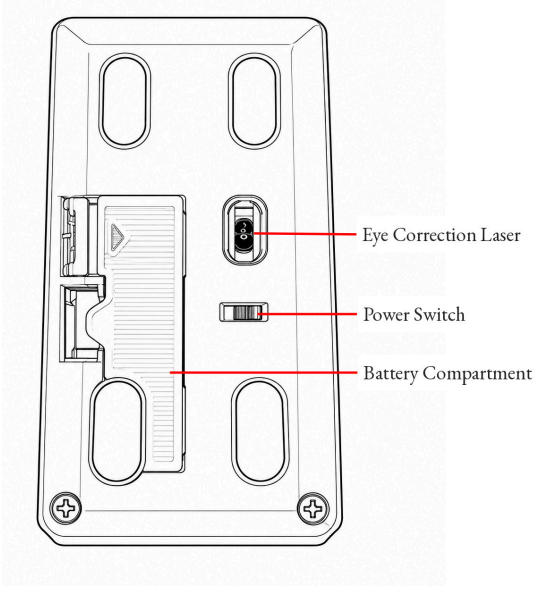
Use the left button for an ordinary click and the wheel for scrolling. The other button is the right button. The illustration doesn't name it, because officially the Macintosh has one button and we have never heard of a right click.

The Sync button is not needed in the normal setup: this mouse is used with its one USB wireless receiver. Leave it alone for ordinary use.



The mouse buttons, scroll wheel, and Sync button. The second button is awaiting official recognition.

Underneath are the battery compartment, power switch, and optical tracking window. The mouse runs on one AA battery. The illustration calls that last item the Eye Correction Laser. It is an optical sensor that tracks the mouse pad. It does nothing for your eyes, so keep it pointed at the desk.



Mouse underside: battery compartment, power switch, and optical tracking sensor.

The mouse can ordinarily stay on between uses, because it drops into a low-power idle state. You can still switch it off for transport or long storage. Its switch has no effect on the keyboard.

USB devices and controllers

Plug the USB hub into the rear USB-C port when you need more than one USB connection, or when a device has an older USB plug. The two game controllers are in the accessory pouch. Chapter 3 explains the game launcher. Use the cables or adapters that came with the controllers, and don't force a plug into the hub.

Bose speakers and stands

The Bose speakers are an optional extra, not part of the PiForma package; they do not fit in the bag. The pair shown are 1984 Bose RoomMate speakers. They are almost as large as the PiForma's screen, which makes for a great deal of sound from a very small picture, and they were chosen because their proportions match.

Connect the speakers to the rear audio output socket, then send the sound there with Sound Options, in the Wi-Fi status menu at the far right of the menu bar. Chapter 3 shows the menu. Give the speakers a stable place beside the working area, with cables out of the way and no strain on the PiForma. Their stands should not make the table or the computer's folding stand unstable.



■ The complete setup: the PiForma on its stand, the keyboard in front, the mouse on its pad, a Bose speaker on its stand at each side and a game controller at the left.

Bringing sound into the PiForma

The external audio adapter brings sound from another device, such as a VCR or a games console, into the PiForma over USB, so it plays through the PiForma's speakers. That makes it external audio into the computer; sound going out uses the rear audio socket, as described above. The rear RCA jack carries only the picture.

Important safety instructions

Keep liquids away, protect the display, and leave ventilation openings clear. Keep the PiForma out of direct sun. Use a stable stand on a level table. Do not carry the computer by cables or press on the screen to move it.

The power switch does not disconnect external charging power from the internal charger. Switching it off doesn't make the inside safe to work on. Never open it up with charging power connected, and leave internal repairs to us.

Stop using the computer if it becomes unusually hot, smells hot, or has damaged power wiring or a damaged battery. Don't keep trying to switch it on. Charge it only with its own power cable or the bag's charging cord. It was built at home, so it carries no safety certification.

Stop using it, too, if the screen starts to flicker. A flicker can have several causes, and it needs our attention before the next session.

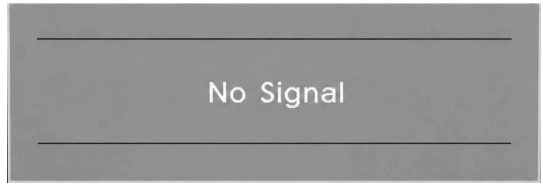
Starting up the computer

The startup chime

A silent startup doesn't mean anything has failed, because some chime positions are silent. Go by the green light and the screen as well.

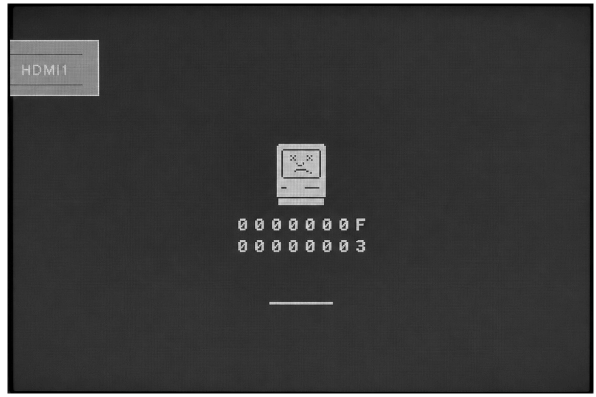
1. **Find the power switch on the right side and switch the PiForma on.**
2. **Look for the green light at the front.**
3. **Listen for a startup chime.** Depending on the rear selector's position, there may be no sound.
4. **Watch for the startup picture or sequence.** The desktop normally follows in about one to two minutes; allow for some variation.

The first thing on the screen is always No Signal. Don't panic: the screen wakes up before the Raspberry Pi does, and there is no way to skip the message. It shows once, the screen goes blank, and the startup picture should appear within 10 to 15 seconds.



Every startup begins with this. It is normal.

The startup picture looks like an old Macintosh that has failed its startup test, sad face, error codes and all. That is our sense of humor. Nothing has crashed and no ROM has failed. The label at the top left names the input the screen is showing, and it should say HDMI.

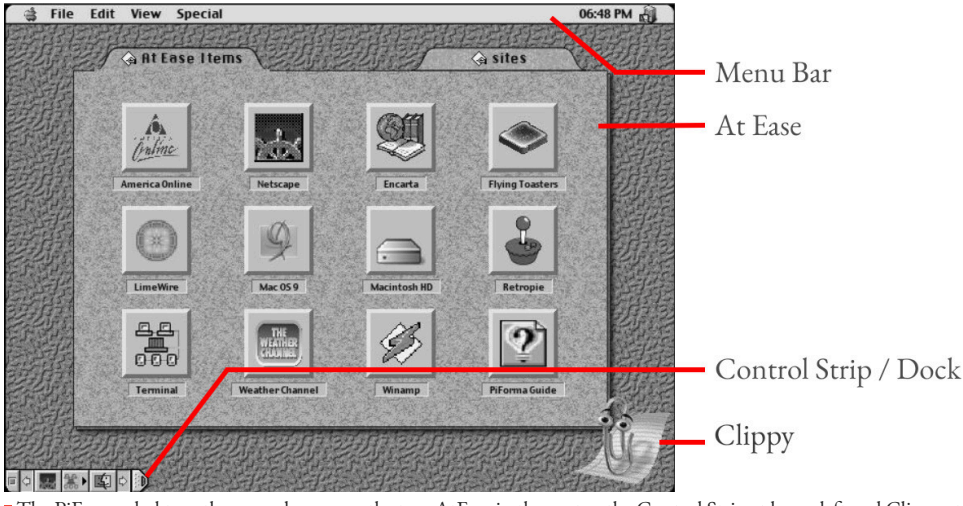


The startup picture. The sad face is a joke; the label at the top left shows the HDMI input.

If the startup picture never appears, press the source selector at the back until the screen shows HDMI. If nothing happens at all when you switch on, turn to “If the PiForma doesn’t start” at the end of this chapter.

Your first session

Once the desktop appears, try this short tour. You need not change any settings or make a file.

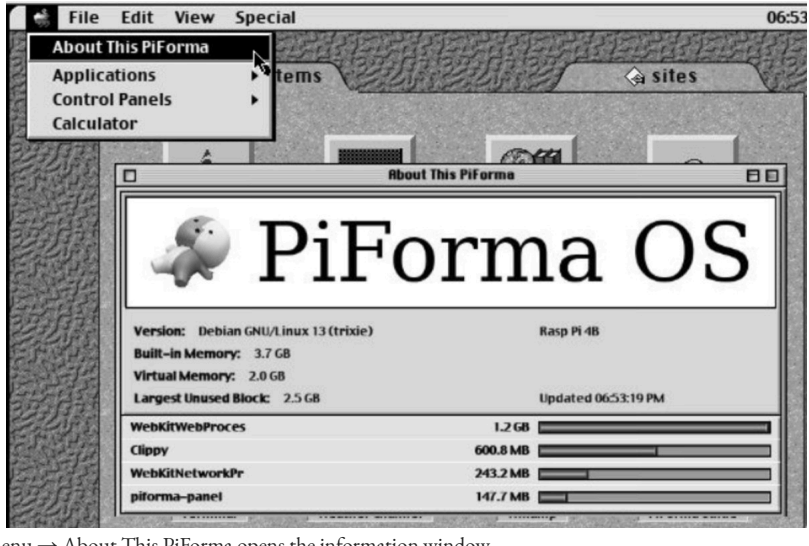


The PiForma desktop: the menu bar across the top, AtEase in the center, the Control Strip at lower left and Clippy at lower right.

1. **Move the mouse on its pad and watch the pointer follow it.** Point at an item by moving the pointer over it. To click, press and release the left mouse button once. The other left button can wait for Chapter 3.
2. **Find the menu bar at the top of the screen and the folder-like AtEase launcher beneath it.** If a window covers the launcher, choose View → Show Desktop from the top menu bar. This minimizes your programs into the Control Strip without quitting them.
3. **In AtEase, click a folder tab if needed, then click a program button once.** AtEase buttons take a single click, not a double-click. Wait for the program's window to appear before clicking again. Use a windowed program for this first exercise; a full-screen game will cover the desktop.

4. **Explore the program's visible controls.** When finished, use the program's own Quit or Exit command if it has one, or its window close control. If asked to save something you want to keep, save it before continuing. Closing one window may leave other windows from the same program open.
5. **Choose View → Show Desktop to return to the launcher.** You can now open another program in the same way.

For a small practice window with a known menu entry, choose Apple menu → About This PiForma. Read the information, close that window using its close control, and return to AtEase. Opening a program, leaving it, and finding your way back are most of the battle. Chapter 3 explains windows, typing, files and games in more detail.



■ Apple menu → About This PiForma opens the information window.

If the launcher or menu bar is missing, see Appendix B rather than trying unfamiliar touch gestures. When you have finished using the computer, follow "Stopping the computer" below.

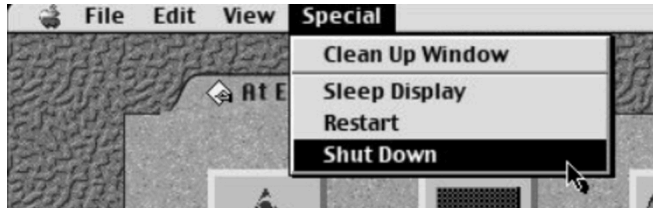
Power for a longer session

If the internal battery is charged, you do not need a power cord for a short session. A full charge typically lasts about six hours of general use, or about four hours of emulated games, especially graphics-intensive ones. For longer sessions, keep the PiForma connected to external power when it is available.

The desktop has no battery gauge, and there is no low-battery warning. Check the battery display through the top slits now and then. When the battery runs out, the PiForma stops at once, just as if the power switch had been turned off, and unsaved work is lost. After a full drain, connect external power and allow a minute or two before it will start.

Take the power cable from the brown pouch. The original rear AC inlet supplies the internal charger. The battery can charge while the computer's power switch is off. Use that cable rather than another charger.

Stopping the computer



Choose Shut Down from the Special menu after saving your work. This view shows the menu choice, not completed shutdown.

1. **Save your work and leave any game or application using its own quit command when available.** If a full-screen program gives you no way out, hold the Apple logo for about six seconds instead.
2. **In the PiForma Panel menu bar, choose Special → Shut Down.**
3. **If a confirmation appears, choose Shut Down.** Let the operating system finish shutting down before removing physical power. A dark screen can also just mean the display is asleep, so give shutdown time to finish.

4. **Once shutdown is complete, turn off the power switch on the right side.** The green light at the front goes out. The wiring permits battery charging while the switch is off.

Never skip that last step. Shut Down stops the software but leaves the power on, and the battery drains quickly until the green light goes out. That matters most in the bag, where you can't see the light.

Switching off without shutting down first is a last resort, for a computer that won't respond. It can lose unsaved work or damage files.

To leave a full-screen program rather than stop the computer, see "Leaving a full-screen program" in Chapter 3. A hold of about six seconds closes only the active application on the current PiForma.

Packing the equipment

1. **Shut down properly, switch off physical power so the green light is out, and disconnect external cables as needed.** Make sure the computer and accessories are cool and dry before putting them away.
2. **Place the PiForma in the bag with its monitor/screen facing upward.** Its rear/bottom must seat in the molded indents in the bottom foam insert.
3. **Gently shift or wiggle it into its intended resting place.** It should not move much when seated. Do not force it down or put pressure on the display.
4. **Switch the keyboard off and check that Caps Lock is up.** Put it beside the computer: keys toward the bag wall, thicker edge toward the rear, thinner edge toward the front.
5. **Place the velvet-covered foam board above the computer.**
6. **Put the mouse pad above the board.**
7. **Put the mouse and collapsible stand above the pad.**
8. **Return the two controllers, hub, and power cable to the brown pouch and place that pouch at the top.** Check that no loose item presses against the screen and close the bag gently.

Charging in the bag

Seating the rear/bottom in the molded foam indents aligns the bag's wireless-charging arrangement. A hidden transmitter sits under the velvet, and its USB cord comes out of the small pocket on the back of the bag. To charge in the bag, plug that cord into a USB charger. This is a slow, trickle-charging arrangement, not fast charging.



The wireless charger's USB cord comes out of the small back pocket. Plug it into a USB charger.

Physical fit does not prove that charging has started. Check the foam seating and that the cord from the back pocket has power. The battery display can show charging from the wireless receiver, but it is hidden while the computer is packed in the bag, so this is rather a hope-and-pray arrangement.

Closing the bag

The bag closes like an old canvas military bag. Feed the strap loop through the four grommets around the top, pass the clasp through the loop, and snap the clasp shut. To open the bag, do the same in reverse.



Pass the clasp through the strap loop and snap it shut.

Carrying the bag

Carry the closed bag carefully. Its padding protects the PiForma during normal handling. Don't drop it. We never crash tested this model, so don't be the first to find out what happens. The Bose speakers travel separately, as described earlier in this chapter.

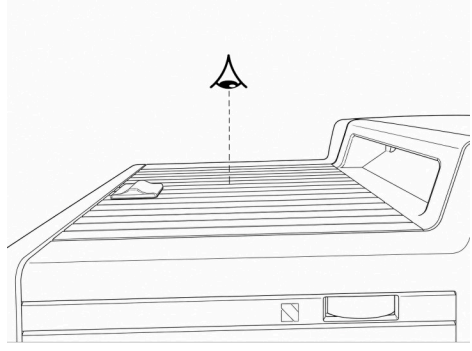
If the PiForma doesn't start

Most startup trouble comes down to the battery, so begin there. Then find the symptom that matches what you see and hear. Appendix B repeats these checks in brief and covers problems after startup.

Checking the battery

If nothing happens, check the battery before assuming something has broken. The outside plunger mechanically presses the button on the internal battery pack, showing its charge indicators; pressing it may also wake or enable the pack.

The normal, management-approved way to read the battery is to press the rear battery plunger to turn on its display, then peer through the slits in the top to see the percentage. You do not have to open the compartment for this check.



Read the battery percentage through the top slits after pressing the rear battery plunger. The eye shows where to look.

Press the plunger, read the battery, and try the power switch again. If the battery is empty, seems asleep, or won't start despite showing charge, plug in the power cable and try again.

If it still won't start, connect the power cable and wait roughly a minute before trying again. We've had this work even when the battery previously showed full. Try it before taking anything apart, but do not count on one minute as a fixed recovery time. We don't know why it works either.

If there is no green light

If there is no green light, no chime, and no picture, start with the power switch, battery indication, plunger and power cable. Then try again. If that doesn't help, don't go looking inside; ask us.

If the light is on but there is no chime

Some positions on the rear chime-selector dial are intentionally silent or empty. With the PiForma powered on, move the dial through several positions in either direction and listen for an available sound.

If it never sounds, the chime module may need attention, which is a job for us. The chime has its own sound hardware, so desktop sound can still work without it.

If the light and chime work but the screen stays blank

No Signal and a moment of blank screen are normal at first. If the startup picture has not appeared after about 15 seconds, press the source selector at the back until the screen shows HDMI. The chime plays before the computer has finished starting, so wait for the picture. If the picture stays blank, flickers or distorts, ask us rather than disturbing anything inside.

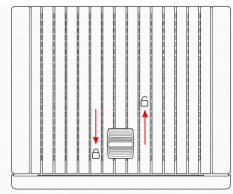
If the desktop is slow to appear

Give the computer one to two minutes to reach the desktop. If it stays on the startup picture, note what you see and tell us rather than switching it off and on repeatedly.

A closer look, strictly off the record

For an optional open-top look, first make sure the vice president isn't looking. He was let go in 1985, but old habits die hard. Then shut down the PiForma, switch off physical power, and disconnect external power. Opening the compartment is for looking, not changing internal wiring.

1. **Slide the top latch toward the open-padlock mark: upward, as the illustration shows.**
2. **With the latch unlocked, lift at the slider to open the upper compartment.**
3. **Close the compartment before sliding the latch toward the closed-padlock mark, downward in the illustration, to lock it again.**



Slide up toward the open padlock to unlock; lift at the slider to open. Close the top before sliding down to lock.

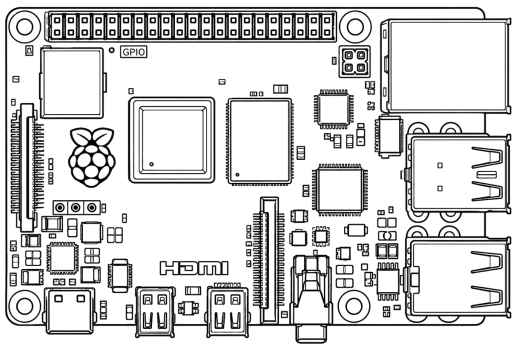
Of course, you did not hear any of this from us. If anyone from management asks, you were merely admiring the ventilation slits. We would appreciate remaining employed.


SERVICEMAN WARNING
THIS PRODUCT INCLUDES CRITICAL ELECTRICAL AND MECHANICAL PARTS FOR SAFETY PROTECTION. FOR CONTINUED SAFETY, REPLACE SAFETY CRITICAL COMPONENT ONLY WITH MANUFACTURER'S RECOMMENDED PARTS. (REFER TO SERVICE MANUAL) FAILURE TO COMPLY MAY BE UNLAWFUL.
OPERATING HIGH VOLTAGE SHOULD BE WITHIN 12.8KV AT ZERO BEAM CURRENT AT 115V AC LINE AS SET FORTH IN SERVICE MANUAL.
TO MEASURE 2ND ANODE VOLTAGE USE HIGH IMPEDANCE METER. CONNECT (—) TO CHASSIS. USE A HIGH VOLTAGE LEAD FROM (+) TO 2ND ANODE.
TO AVOID POSSIBLE EXPOSURE TO X RADIATION, TAKE X-RADIATION PROTECTIVE MEASURES(SEE SERVICE MANUAL) FOR PERSONNEL DURING SERVICING.
DISCHARGE HIGH VOLTAGE TO B MINUS ONLY. ARCING TO ISOLATED METAL PARTS PRODUCES A SHOCK HAZARD.
THE PICTURE TUBE IN THIS RECEIVER EMPLOYS INTEGRAL IMPLOSION PROTECTION. REPLACE WITH A TUBE OF THE SAME TYPE NUMBER FOR CONTINUED SAFETY.
HIGH VACUUM PICTURE TUBE IS DANGEROUS TO HANDLE. REFER SERVICING TO QUALIFIED SERVICE PERSONNEL.
4549471

CHAPTER 2

Inside the Macintosh PiForma

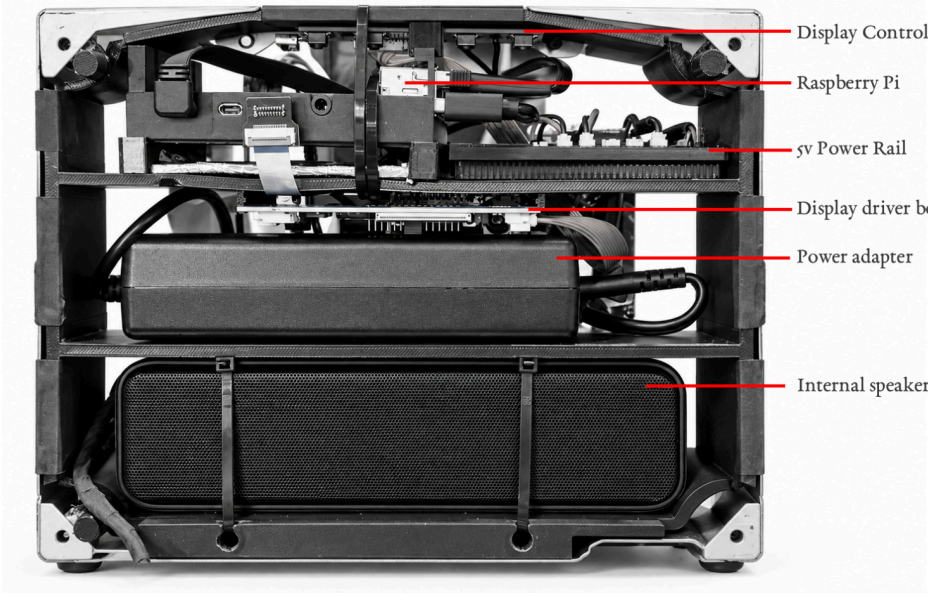
The PiForma looks like a 1985 monitor, but the computer inside is entirely modern. This chapter introduces the parts and what they do. Appendix A has the finer details for anyone who wants the wiring.



The monitor that became a computer

The case began as an Apple IIc Monitor. Its original CRT had been removed before this build. The plastic case, metal chassis, power switch, AC inlet, and rear openings were reused; the computer and screen inside are new.

Nothing here was restored to factory condition. The old case was a good place to start, and a difficult place to fit a computer, a screen and a battery.



The internal assembly, with the Raspberry Pi, display hardware, power adapter, 5V power rail and ordinary internal speaker identified.

Raspberry Pi, memory and storage

microSD: The small storage card that holds the operating system, the programs and your files.

A Raspberry Pi runs the programs. RAM holds the work the computer is doing now; the microSD storage card keeps the operating system, applications, and saved files when power is off. Saving transfers your work into a file that can remain after shutdown.

The computer runs Raspberry Pi OS beneath its Macintosh-inspired appearance. Appendix A lists the hardware and software specifications.

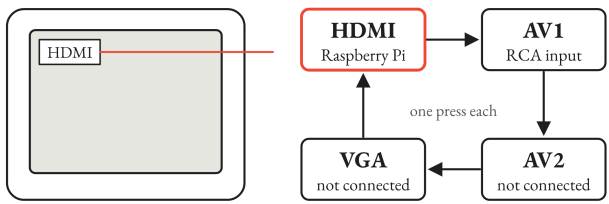
The display and source selector

Composite video: The older video signal used by VCRs and early game consoles. The RCA jack carries picture only, not sound.

An LCD screen replaces the original monitor display. The Raspberry Pi supplies its picture over HDMI. A rear RCA input supplies composite video from an external source: a VCR, an older games console or anything else with a composite video output. Its sound comes in separately, through the external audio adapter.

The picture quality isn't particularly good. The screen is a little dim, the colors aren't quite right, and neither brightness nor color can be adjusted from the computer. We spent the display budget elsewhere.

Each press of the rear source selector moves the screen to its next input. Only two are in use: HDMI, for the Raspberry Pi, and AV1, for the composite RCA input. The selector also passes AV2 and VGA, which had nothing connected when the PiForma was built. A small label at the top left of the screen names the current input. Press until it shows HDMI to return to the Pi's picture.

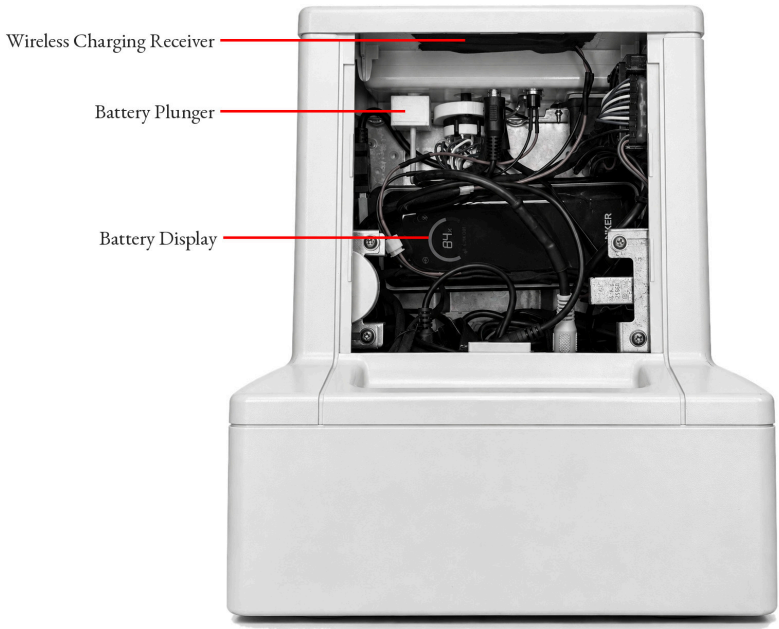


Each press of the source selector moves the screen to its next input. HDMI shows the Raspberry Pi; AV1 shows the RCA input.

The battery and charging paths

The internal battery lets you use the PiForma without a power cord. The power cable feeds its charger, and the battery can charge while the computer's physical switch is off. The switch does not disconnect the charger from external power. Appendix A describes the internal power path.

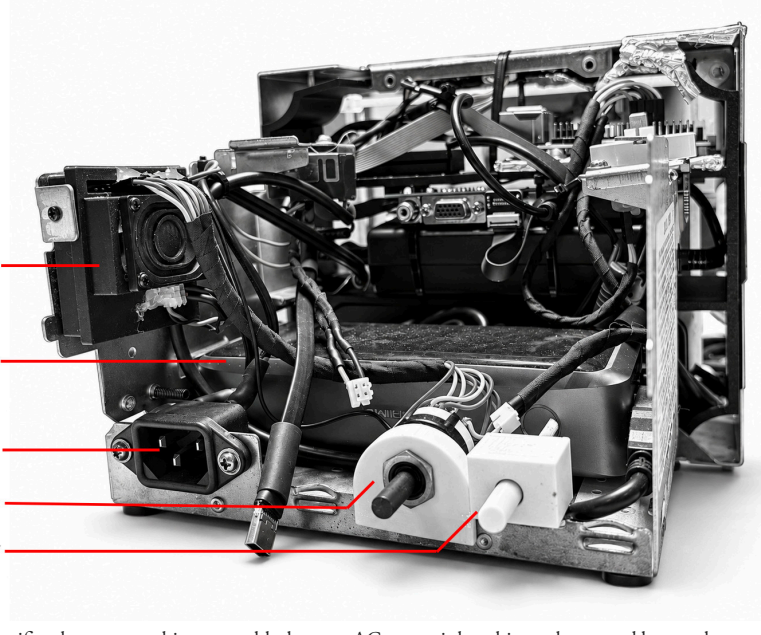
An external plunger reaches the battery's own button. Press it to check the charge, and sometimes to wake the pack. The normal check is to press the rear plunger and read the percentage through the top slits. The top latch can also be unlocked and lifted for a closer look; Chapter 1 describes the optional opening procedure.



The open top reveals the wireless charging receiver, battery plunger and battery display. The battery is at 84 percent in this photograph. For the normal check, read through the top slits as described in Chapter 1.

System sound and startup chimes

The startup chime and ordinary system audio are separate. Ordinary sound plays through the internal speaker behind the screen or, when chosen in Sound Options, through the rear audio output socket. A separate sound module and speaker play the startup chime; the rear selector chooses among its sounds. Some positions are silent or empty.



A rear oblique view identifies the separate chime assembly, battery, AC power inlet, chime selector and battery-button plunger.

Startup chimes

Turning the selector one click at a time gives these sounds, in order:

1. The classic Macintosh startup chime.
2. An older Macintosh chime.
3. A Macintosh Plus floppy drive working at startup.
4. The Game Boy Advance startup sound.
5. The Apple II beep.
6. The PlayStation 2 startup sound.
7. A picture tube warming up.
8. The Windows 98 startup sound.
9. The two-note Windows 3.1 fanfare.

The next three clicks are silent, and the one after that brings back the classic chime.

Normal desktop sound uses an internal USB speaker. Its volume has nothing to do with the startup chime, which has its own hardware.

The physical controls

The right-side power switch controls the computer's power rail. The Apple logo on the front is a touch control for shortcuts. The rear source selector chooses the LCD input, while the rear chime dial chooses the startup sound.

Turn the side volume knob clockwise to raise normal system volume and counterclockwise to lower it. Pressing the knob does nothing; it has no mute.

Ports, wireless links and adapters

One rear USB-C port is available to the user. The Raspberry Pi's own ports are inside the case and out of reach. Use the hub for several devices, or for devices with older USB plugs.

Bluetooth LE: Bluetooth Low Energy, the wireless link the converted keyboard uses.

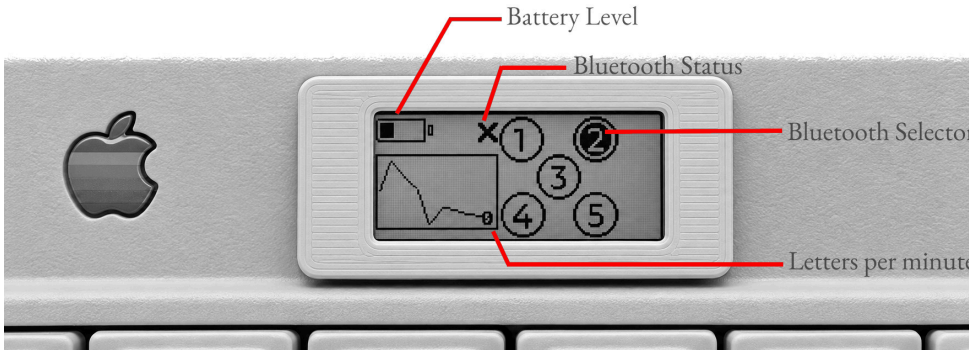
The keyboard uses Bluetooth LE and the mouse uses its USB wireless receiver, which is plugged into the Raspberry Pi inside the case and stays there. Wi-Fi and Bluetooth are handled by the operating system. External audio input uses a separate USB capture adapter and audio-loop helper, not the rear composite-video RCA jack.

The converted Apple keyboard

The vintage Apple keyboard has been converted to Bluetooth LE using Matt Chesters' keyboard-conversion project. It has its own battery, a nice!view display, a volume dial on its left-hand edge, and a power switch and USB-C charging port on its right-hand edge. Appendix A and the credits identify the conversion and its source.

The keyboard has been painted, not retrobrighted. Old Apple keyboards yellow with age. Today they are often bleached back to their original color, a process called retrobrighting, but in the old days the fix was a coat of paint in the nearest matching color. That is the source of the slight orange-peel texture you can feel on the case.

The nice!view display, above the number keys, shows four things. The battery level is the keyboard's own charge. The Bluetooth status shows an X when the keyboard isn't connected, as in the photograph, where the PiForma was switched off; once connected, it shows a wireless symbol. The Bluetooth selector shows which of five Bluetooth profiles is in use, and the PiForma is profile 2. The graph shows your typing speed in letters per minute, not words.



The nice!view display: battery level, Bluetooth status, Bluetooth selector and letters per minute.

The keyboard's small 100 mAh battery lasts a couple of weeks of being switched on with light use. When the battery level runs low, charge it from the USB-C port on the right-hand edge.

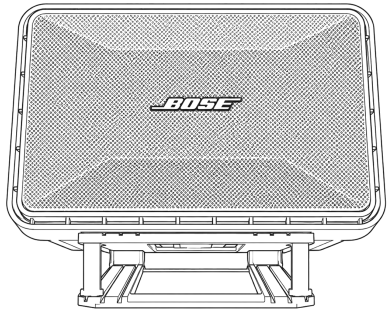
Pressing down on the display changes the Bluetooth profile, one step per press; press on its plastic surround rather than the screen. Leave it alone: there is no reason to press it unless you want the keyboard to talk to a different device, and it should stay on profile 2 for the PiForma. The 3D-printed key around the display is a little too wide for its opening, so it tends to stick down when pressed. If it does, lift it back up with a fingernail.

The matching mouse

The matching mouse combines Bambu wireless mouse-kit electronics with a printed Retro Wireless Computer Mouse shell. The scroll wheel and axle were replaced after the original wheel broke during modification. A USB wireless receiver provides the connection to the computer.

Bose speakers and stands

The Bose speakers are optional; the PiForma's internal speaker works without them. Chapter 1 explains connecting and placing them.



A Bose RoomMate speaker on its stand. The pair are optional, and travel outside the bag.

The bag and its fitted interior

The PiForma travels in a waxed canvas backpack, finished with Otter Wax, so it shrugs off the weather and only looks better with age. Colorful embroidery and a set of patches give it a personality all its own. We are known for minimalist design. Here, we went a little rogue.

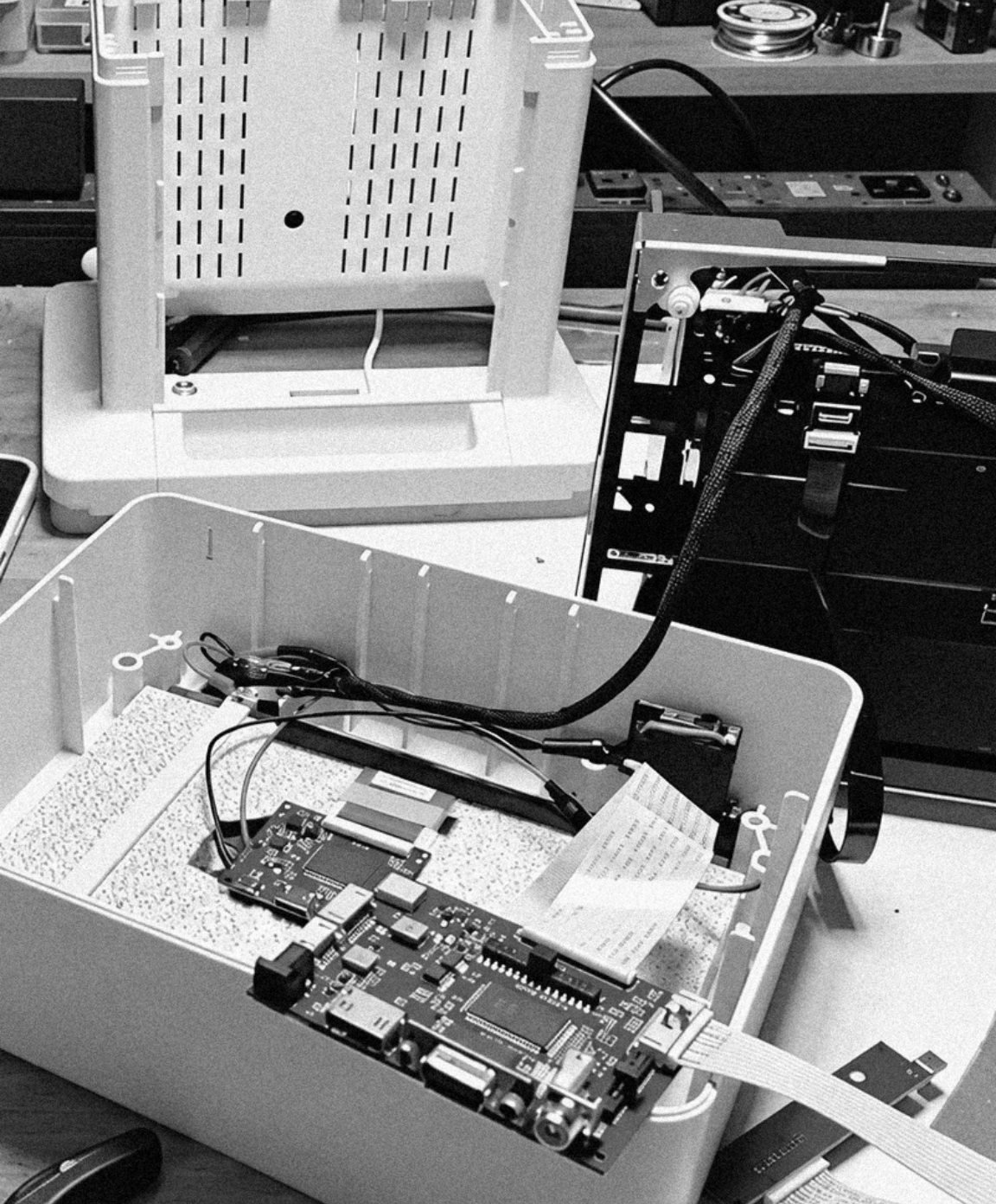
Inside, smooth green velvet holds the computer as if the two were made for each other, because they were. A corrugated-plastic liner gives the bag its shape, a foam bottom cushions it, and alignment grooves set the PiForma in exactly the right place. A wireless charging transmitter is concealed under the lining, with its cable reaching the back pocket, so the computer charges while it rests.

The foam locates the computer as well as cushioning it. The packing sequence in Chapter 1 protects the screen and puts the keyboard in its intended side position. A computer that only roughly fits won't charge; it has to sit in the grooves.

It looks right on anyone, and especially right over a tweed jacket between business trips. There is no room left for clothes. You won't miss them. You'll be the envy of the departure lounge.



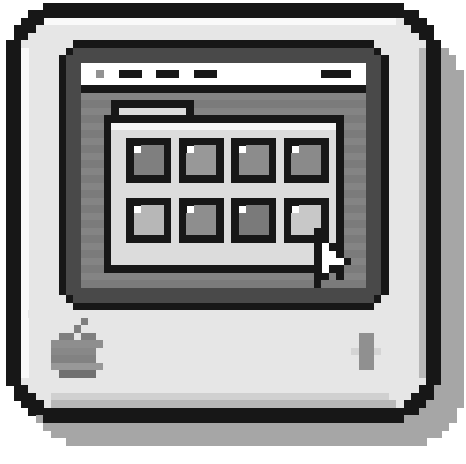
Waxed canvas, embroidered, and fully loaded. The strap loop runs through four grommets and the clasp holds it closed.



CHAPTER 3

How the Macintosh PiForma Works

The PiForma looks like a Macintosh, but Raspberry Pi OS does the work underneath. This chapter explains the everyday business of pointing, typing, opening programs and keeping what you make. You need not learn the wiring to save a file.



Finding your way around the desktop

AtEase: The folder-like launcher that opens programs with a single click.

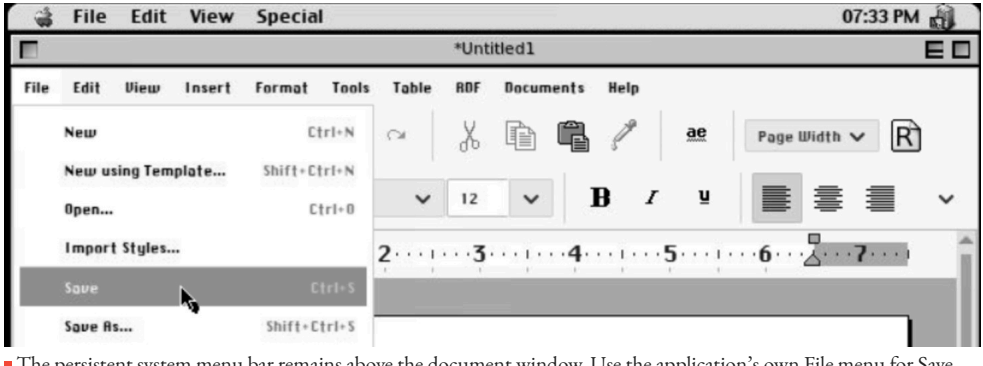
The desktop is the working screen behind your application windows. PiForma Panel supplies the menu bar at the top; AtEase is the folder-like launcher for programs. The Control Strip provides another way to open or return to programs.

The desktop picture in Chapter 1 shows these parts together. The launcher has At Ease Items and sites tabs; Clippy occupies the lower-right corner. The clock appears at the upper right of the menu bar.

Choose a menu by clicking its name, then click the command you want. An unavailable command may be dimmed. Read a dialog before answering: it may ask where to save a file, whether to discard changes, or whether to stop the computer.

Two menu bars, two different jobs

This is where the PiForma differs from a real Macintosh. On a Macintosh, the menu bar at the top of the screen belongs to whichever program is in front, and its menus change when you switch programs. **On the PiForma, the menu bar at the top never changes.** It is a system menu bar, and its commands are the same whichever program you are using.



The persistent system menu bar remains above the document window. Use the application's own File menu for Save and its other document commands. The themed window's close control is at upper left; minimize and maximize are at upper right.

Each program keeps its own menus inside its own window. For document work, use that program's own File → Save, Edit → Copy, Edit → Paste, and other available commands. The top File menu and a document window's File menu have different jobs, so be sure you are opening the one you mean. It looks like a Macintosh, but not every classic Macintosh command works; use the commands each program shows.

Using the mouse

Move the mouse across its pad to move the pointer. If you reach the edge, lift the mouse, set it down nearer the center and continue. The mouse need not point at the screen; the little eye underneath follows the surface below it.

Click means press and release the left button once. A click selects an item or operates a button. Double-click means two quick clicks without moving the pointer between them; many file views use it to open an item. AtEase program buttons need only one click. Repeated clicking while a program starts can produce more windows than you intended.

To drag, point at the object, hold the left button, move the mouse, then release. Drag a window by its title bar, rather than by its contents. Dragging a file may move or copy it, so practice with windows before rearranging your work.

Roll the wheel to scroll through a longer page or list. Point over the area you want to scroll. Scroll bars provide another way to reach material outside a window's visible area.

The right button may show a menu of actions for the item under the pointer. Its behavior depends on the program. This is the mysterious "??????" in the Chapter 1 illustration; it is a real right button, despite our best efforts.

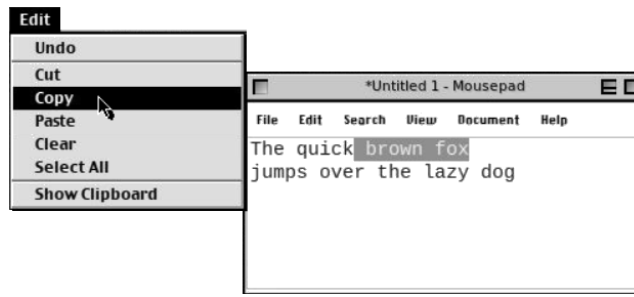
Using the keyboard

Switch the keyboard on at its right-hand edge. First click in the window or text field where you want to type. The insertion point marks where new text will go; moving the mouse pointer alone does not move it.

Use the letter, number and punctuation keys to enter text. Hold Shift while typing a letter for a capital or a key's shifted symbol. Return commonly ends a paragraph in a document; in a dialog it may accept the current choice, so read the dialog first. Use the arrow keys to move through text where the program supports them.

To replace text, select it by dragging across it, then type the replacement. Deleting selected text removes the whole selection. If you remove something by mistake, look for Undo in that application's Edit menu before doing more work.

Use an application's own Cut, Copy and Paste commands when available. Copy leaves the original selection in place; Cut removes it; Paste inserts the copied or cut material at the insertion point. Check the selection before cutting, and the insertion point before pasting.



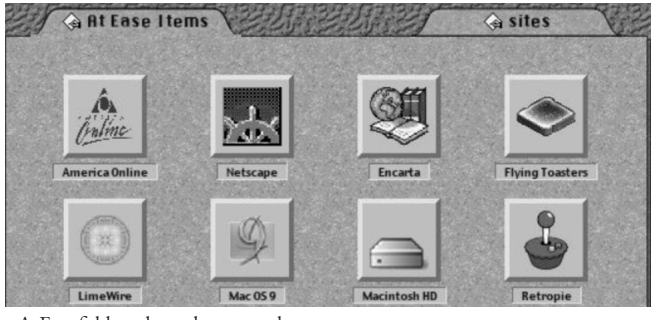
Selected text is highlighted. Choose Copy or Cut from the Edit menu, click to place the insertion point, then choose Paste.

The vintage keys don't map neatly onto modern shortcuts, and the Apple key may not do what it did in 1985. Use the commands each program shows in its menus. If ordinary typing fails, use the keyboard checks in Appendix B.

Opening programs with AtEase

AtEase

AtEase has two folder tabs. At Ease Items holds the programs; sites holds a selection of websites we find interesting, a collection still being tidied. Each tab shows only what fits in its window; there is no second page yet. Click a tab to switch between them, then click a button once. Allow time for the program to open. A launcher button starts a program, not a document.



AtEase folder tabs and program buttons.

Show Desktop

If application windows cover the launcher, choose View → Show Desktop from PiForma Panel. This minimizes your programs into the Control Strip without quitting them. You can then select another AtEase button. If the launcher is absent rather than covered, see Appendix B.



View → Show Desktop returns you to the launcher without quitting the open applications.

Working with application windows

Several programs can be open at once. Click a window to bring it forward and direct your typing to it. Before typing a command or entering text, check that the intended window is active. An overlapping window is still open even when you can see only an edge of it.

Where a normal window has a title bar, drag it to a more useful position. Where a resize border or corner is provided, drag that border or corner to change the window's size. Some programs fix their window size or use the whole screen; those controls will not always be present.

Use a program's Quit or Exit command to leave it. A window close control closes that window, which may leave other windows from the same program running. A save question gives you a chance to keep changes; Cancel returns you to the work so you can decide.

Showing the desktop minimizes your programs into the Control Strip without saving or quitting them. Return to an open window by clicking its visible part or the appropriate Control Strip pane. Give the strip a moment to detect a newly opened program. Return to a running program from the Control Strip rather than opening it again from AtEase.

Window controls

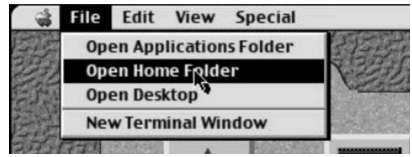
The themed document window in the menu-bar picture, near the start of this chapter, has a close control at the upper left. At the upper right, the far-right control maximizes the window; the control immediately to its left minimizes it to the Control Strip. Minimizing puts the window out of the way while leaving it open. Use the program's Control Strip pane to return to it.

Use the controls a window actually provides. Some windows omit controls or draw them differently. Clippy's chat, for example, has its close button at the upper right. Closing a window is not the same as minimizing it; save before closing work you want to keep.

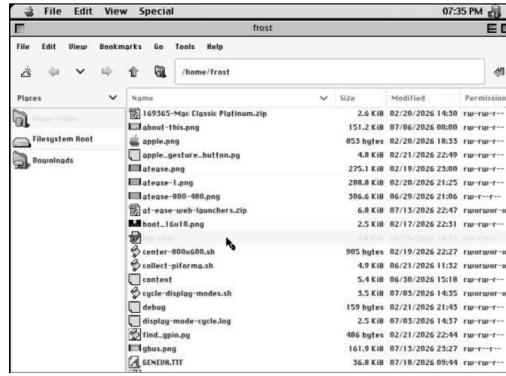
Files and folders

A file holds something the computer can keep: a document, picture, sound or other information. A folder groups files and other folders. A program is the tool you use; the document is the work you keep. Closing the tool does not substitute for saving the work.

PiForma Panel's File → Open Home Folder opens your home folder in a file window. Browse by opening the folders shown there. Many file views open a folder or file with a double-click; use the view's Open command if its click setting differs. Opening a document starts the program that goes with it; an unfamiliar file may ask which program to use.



Use the system File menu's Open Home Folder command to reach your files.



The home-folder window lists files and folders and has its own menu bar. The contents shown belong to this captured session; your list may differ.

Keep related work in a folder you can recognize. Give documents meaningful names and note which folder contains them. Before moving, renaming or deleting a file, close it in its program and make sure you know which item is selected. Leave unfamiliar system and application files alone.

Moving or copying a file doesn't save the changes in an open document; only Save does that.

Backups

The PiForma has no backup arrangement for its microSD card, and management prefers not to discuss it. The build is documented well enough to rebuild from scratch if the card ever fails, but your own files would not survive that. Two copies on the same card do not protect against failure of that card.

Copy any work you value to a USB drive or another computer.

Saving and reopening your work

Programs do not all save in the same way. Use this sequence in a document application that offers Save and Open commands; a web page or game may use a different system.

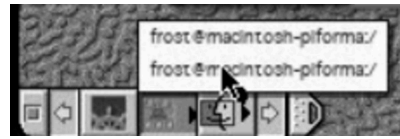
1. **Choose Save from the application's own menu.** On the first save, it will usually ask for a name and location.
2. **Select a folder you can find again, enter a useful filename, and confirm with the dialog's Save control.** Keep the file type the program suggests.
3. **If warned that a file already exists, check its name and folder.** Replacing it may overwrite earlier work. Cancel and choose a different name if you want both versions.
4. **After further changes, use Save again.** Where offered, Save As lets you choose another name or location; read the dialog to see which copy you will be editing afterward.
5. **To return later, use the application's Open command and find the saved file, or open the file from its folder.** Check that the expected work appears.

Save often, and always before quitting or shutting down. If a save fails or a dialog is still waiting, deal with it before closing. Don't count on any program saving for you.

The menu bar and Control Strip

Apple menu → About This PiForma opens system information. View → Show Desktop uncovers the launcher. Special → Shut Down requests an operating-system shutdown; Special → Sleep Display only blanks or sleeps the display. A sleeping screen is not a stopped computer.

The Control Strip sits at the bottom-left of the desktop. The original Macintosh Control Strip held control panels; the PiForma's works more like a dock. It holds a pane for every program you have open, and for programs kept there for quick access. Click a program's pane to launch it when closed or return to its windows when open.



The Control Strip at lower left, with a window popup visible above it. The rest of the desktop is omitted so the strip can be read clearly.

Apple-logo shortcuts

The Apple logo on the front is a touch control. Besides the six-second hold described under "Leaving a full-screen program," short taps open programs directly. Leave about half a second between taps.

Each touch blinks the green light at the front off and on again, so you can see that the PiForma felt it: one tap, one blink; two taps, two blinks. Keep objects away from the logo, or the sensor may take them for a touch.

Apple logo	Opens
Tap once	The browser
Tap twice	Mac OS 9
Tap three times	Winamp
Tap four times	RetroPie
Hold about six seconds	Closes the active application

Controllers and games

The accessory pouch holds two game controllers. Plug a controller into the rear USB port, or both into the hub. If a controller has an older plug, use the adapter that came with it, and don't force anything.

To play games without unpacking the keyboard and mouse, switch on, connect a game controller, wait for the desktop and tap the Apple logo four times to open RetroPie.



Two game controllers and a racing game on the PiForma screen.

EmulationStation: RetroPie's game-list menu. It is separate from the games it starts.

Open RetroPie/EmulationStation from AtEase to reach the game collection. Use the controls shown by the launcher to choose a system and game. Follow the game's own instructions for play. Buttons can do different things in different games, so the A button won't always mean the same thing.



EmulationStation's game list. The bar along the bottom names the buttons for this screen, including A to launch the highlighted game and B to go back.

Before settling into a game, find out how that game pauses, saves and exits. The controllers are already set up, so if a screen asks you to set one up, read it carefully rather than redoing the setup just to make it go away. If you can't work out the controls, ask us.

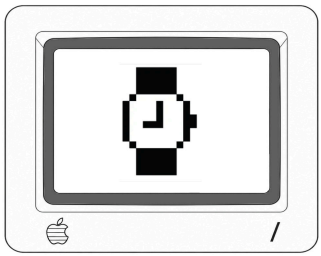
Save state: An emulator's snapshot of a game in progress, separate from the game's own save.

A game can save in two ways: its own save, and the emulator's save state, a snapshot of the moment. Use whichever the game offers and check that it worked. Closing a game with the Apple logo does not save it.

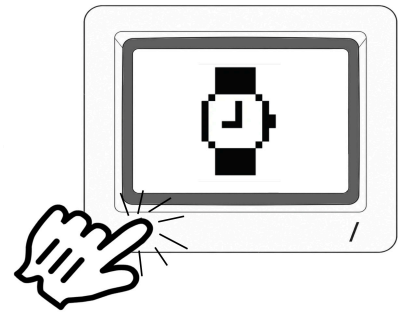
Leaving a full-screen program

A full-screen game can cover the menu bar and window controls. First use its own Quit or Exit command when available. Save before leaving whenever the program allows it.

Holding the Apple-logo touch control for about six seconds closes only the active application. Use it when a full-screen program gives you no way out. It closes one program, not the computer. Unsaved work may be lost.



An application has taken up the full screen with no way to exit.



Touch and hold the Apple logo for about six seconds to close the active application.

The same hold is how you quit a game. The game controllers use every button, so none is set aside for quitting. Hold the Apple logo for about six seconds: the game closes and EmulationStation returns. To leave EmulationStation too, hold the logo for another six seconds; EmulationStation closes and the desktop returns.

Emulator: A program that imitates another computer or games system so its software can run.

Some emulators trap the mouse pointer inside their window. If you can't get it back out, use the six-second hold, or see Appendix B.

Diversions

The project includes a browser and a collection of themed diversions, with Clippy, Weather Channel, Flying Toasters and music-player experiments among them. A familiar old name may open a modern program dressed up to look like the original software.

The browser is Chromium, dressed as Netscape Navigator; its AtEase button carries the Netscape name. Type an address in the address field to visit a site. Online pages need Wi-Fi; programs stored on the PiForma itself work without it.

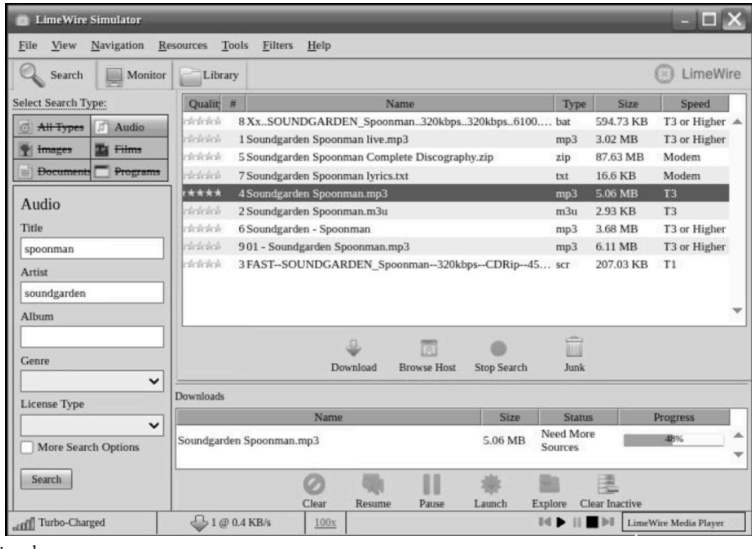
The other AtEase buttons are mostly what they seem. America Online opens an America Online simulator, and Encarta opens Kiwix, a reader for offline reference articles, at least as of this writing. Macintosh HD opens a file window. Mac OS 9 runs the real Mac OS 9 inside the SheepShaver emulator; it fills the screen, so leave it with the six-second hold. Terminal opens a command line, which is for us.

Winamp is the QMMP music player wearing a Winamp skin. It plays the songs in the Music folder in your home folder; to add more, copy them into that folder.



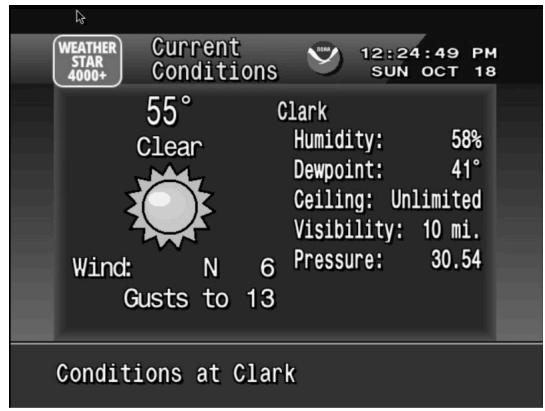
Winamp and its playlist window open over the AtEase launcher.

LimeWire Simulator is a period-styled take on the old file-sharing program.



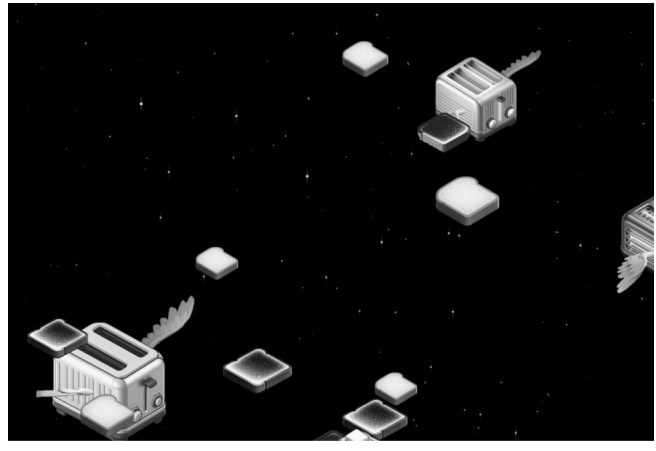
LimeWire Simulator.

The Weather Channel estimates your location from your internet connection and shows the nearest weather station.



The Weather Channel diversion showing current conditions in WeatherStar 4000+ style.

The Flying Toasters are a screen saver you start yourself. Because the animation works the computer hard, they never start on their own when the computer is idle. To stop them, use the six-second hold.



The Flying Toasters, airborne.

Clippy

Clippy is the assistant at the lower-right corner of the desktop. In the Chat with Clippy window, click the message field, type your question, then press Enter or click Send. The field itself reminds you how to send. Read the reply in the conversation area; use its scroll bar to revisit a longer conversation.

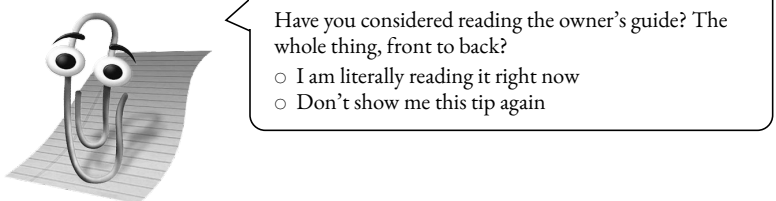


Enter a question in the message field, then press Enter or click Send.

Clippy can offer suggestions, but check them against the PiForma's actual setup before changing settings or installing anything. He is very enthusiastic, and he is not always right. We have given him his own corner of the screen.



An example Chat with Clippy conversation. Its suggestions should be checked against the actual PiForma setup.



About This PiForma and the project guide

About This PiForma displays computer and operating-system information. Open it from the Apple menu.

PiForma Guide is the built-in guide to the computer and its functions, with the project's hardware, software, maintenance and other documents. Its Update Stack button checks for updated documentation. If an update fails, the guide keeps its previous copy; without one it uses its bundled offline snapshot. An offline copy may be out of date if the PiForma has changed since.

Click PiForma Guide in AtEase; its window is titled Macintosh PiForma Guided Tour. Choose a stack such as Meet PiForma, Hardware or Software to begin reading. The bottom line reports documentation status and identifies the snapshot; it is worth checking when the machine and a document seem to disagree.

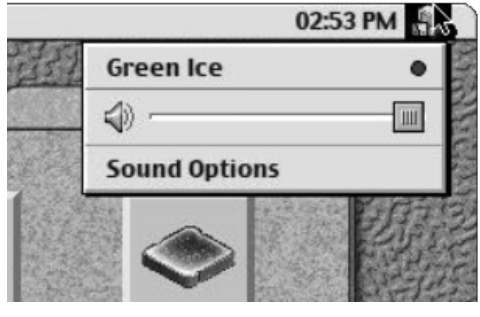


PiForma Guide opens Macintosh PiForma Guided Tour. Choose a documentation stack; the bottom line shows documentation status and the snapshot information.

Wi-Fi and Bluetooth

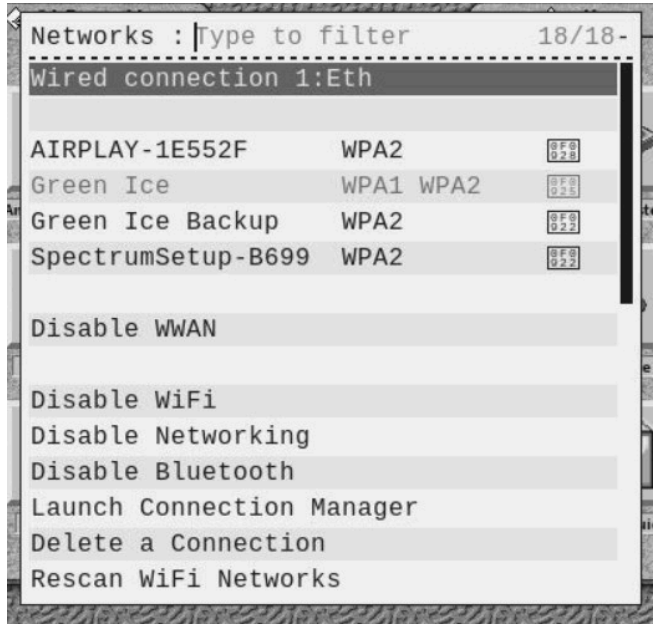
Wi-Fi connects the computer to a wireless network. Bluetooth connects devices such as the keyboard. The mouse's USB wireless receiver is a third, separate connection. So if a website won't load, the keyboard and mouse aren't to blame, and if the keyboard stops, Wi-Fi isn't either.

The Wi-Fi status icon is at the far right of the menu bar. Click it and wait: the menu can take 20 or 30 seconds to appear while it checks the wireless connection. It does this every time. The first line names the network the PiForma has joined, Green Ice in the picture, and a green status means it has joined, though not necessarily that the internet can be reached. The slider below it sets the volume, and Sound Options at the bottom chooses where the sound goes.



The Wi-Fi status menu: the network the PiForma has joined, the volume slider and Sound Options.

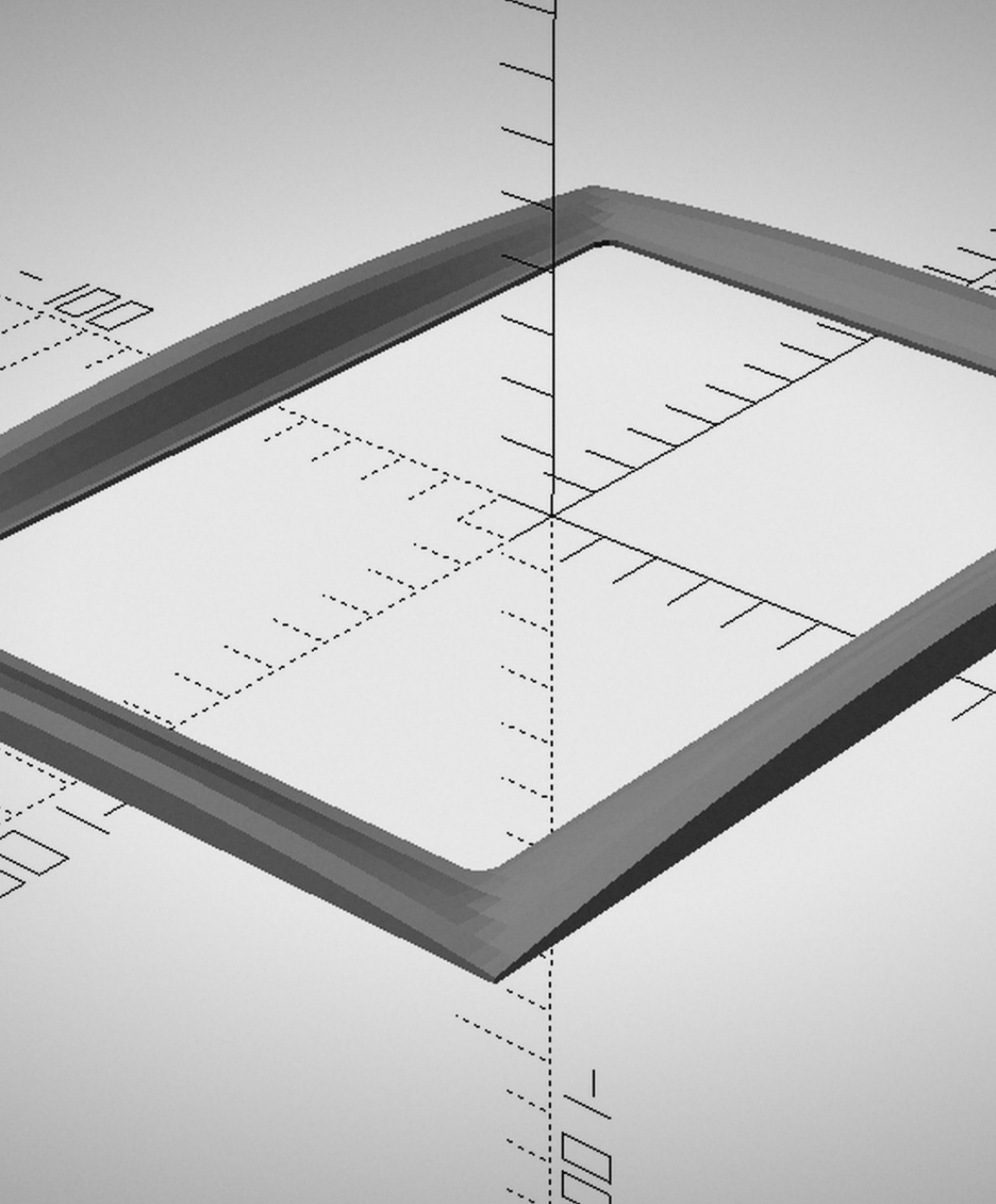
To join a different network, click the network's name at the top of the menu. A wireless network manager opens. Give it a few seconds while it scans for nearby wireless signals, whatever those are; it is 1985, after all, and wireless networking has not been invented. Then click the name of the network you want to join. If a connection fails, see Appendix B.



The wireless network manager after its scan. Click the network you want to join.

Choosing where sound goes

Sound Options, at the bottom of the same menu, opens a control panel for changing the sound output. Choose the internal speaker, or the rear audio socket for speakers or headphones such as the Bose speakers in Chapter 1.



CHAPTER 4

Caring for Your Macintosh PiForma

The PiForma was built to be used and carried, with a little care. Its case has already had a long life; the electronics inside are newer and more delicate.



Handling the computer and display

Use the carrying handle as described in Chapter 1 and watch the keyboard as the computer comes out of the bag. Avoid impacts, pressure on the display, and pulling on connected cables.

Don't try to cure a flicker by tapping the case or poking inside; tell us.

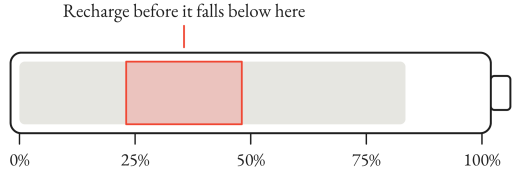
Keyboard and mouse care

Keep crumbs, liquids, and loose items out of the keys. Switch the keyboard off before packing it, and check that Caps Lock is up (Chapter 1 explains why). Turn the mouse off for prolonged storage or transport if desired.

The keyboard's nice!view display shows its battery level, as Chapter 2 explains. Charge it from the USB-C port on its right-hand edge when the level runs low.

Charging and storage

We prefer to keep the main battery above roughly 25–50% most of the time rather than repeatedly running it empty. That's our habit. The battery maker never asked us to.



Our habit: recharge before the battery display falls below roughly 25 to 50 percent.

Use the PiForma's own power cable. A battery that looks full may still need the plunger or a minute on power before it starts, as Chapter 1 describes. Bag charging requires the foam alignment and a USB charger on the cord from the bag's back pocket. You can't see the battery display in the bag; Chapter 1 explains. Always switch the PiForma off before it goes into the bag.

Stand, bag and transport

Put the folding stand on a solid flat table, open it securely, and center the PiForma. The stand may permit some side-to-side motion, so keep it away from bumps and do not use it on an unstable surface.

Pack in the exact order in Chapter 1. Seat the computer in the bottom foam indents, orient the keyboard correctly, and replace the protective board before the pad, mouse/stand, and pouch. Fasten the clasp as Chapter 1 shows. The bag is padded. Don't drop it anyway.

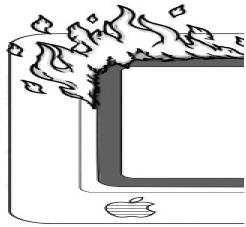
Bose speakers and stands

The speakers travel separately from the bag. Move the speakers and stands carefully and keep their cables from pulling on the computer.

Cleaning, ventilation and unusual behavior

Switch off and disconnect charging power before ordinary exterior cleaning. Use a soft dry cloth gently on the case; keep liquids and solvents away from openings and internal parts.

To clean the screen, wipe it gently with a washcloth that is only very slightly damp, wrung out until no water drips. No soap is needed.



Warmth gathers at the top left, where the Raspberry Pi sits. Warm is normal. This is not.

Leave vents clear and keep the PiForma out of direct sun, which will toast it. Watch for unusual heat, smell, swelling, damaged wiring, a flickering screen, or electrical behavior that is out of character. Stop using the machine in that condition. A desktop that still appears is no reason to ignore any of these.

The Raspberry Pi sits behind the top left of the screen. To check its temperature, rest a hand on the top left of the case. Warm is normal; if it feels too hot, shut down and switch off. The Pi protects itself by slowing down when it gets very hot, but it doesn't switch itself off, so it is up to you.

Service and support

There isn't any. Sorry.

Only one Macintosh PiForma was ever made, and it was never offered for sale. There is no service department, warranty desk, or helpful person in a tie waiting by the phone. If it needs repair, it will probably come back to us, assuming we are in the mood.

That does not make unusual heat or electrical damage a joke. Stop using an unsafe-looking machine and speak to us rather than continuing to experiment. Treat it kindly, and with any luck neither of us will have to find out how much paperwork the service department requires.



The service department, standing by.

Appendix A: Technical Information

This appendix describes the PiForma's hardware, connections and software. Use the project documentation for maintenance and recovery instructions.

Hardware inventory and identification

Donor:	Apple IIc Monitor A2M4090, February 1985.
Computer:	Raspberry Pi 4 Model B, 4 GB RAM.
Storage:	256 GB microSD.
Display:	9-inch LCD kit, working mode 800 × 480; about 640 × 480 shows inside the bezel, a little more across. Software is sized for 640 × 480. Brightness and color are fixed.
Size:	the original monitor's exterior, unchanged.
Weight:	8 lb 13 oz (4.0 kg), close to the original monitor.
Battery runtime:	typically about six hours, or about four hours of emulated games, especially graphics-intensive ones.
Main battery:	Anker Prime 26K 300W.
Internal charger:	100W USB-C adapter.
Keyboard:	Apple A9M0330 ADB keyboard, painted, BLE conversion, ZMK, 100 mAh battery, USB-C charging, nice!view, volume dial.
Mouse:	Bambu kit electronics in a printed shell, USB wireless receiver.
Startup sound:	YS-M3, CK1028 selector, separate salvaged speaker.
Normal sound:	internal USB speaker, or the rear audio output socket for speakers or headphones.

Power architecture

Power path: rear AC inlet → internal charger → main battery → original power switch → custom 5V JST rail → Pi, LCD controller, startup-chime module, and touch sensor. The USB speaker is supplied through the Pi. The switch does not disconnect the AC charging path. The rail has no separate installed fuse protection.

Connections and GPIO

Apple-logo sensor:	TTP223 on BCM GPIO23, active-low.
EC11 volume encoder:	BCM GPIO17/27; push input GPIO22 exists in software but is not physically usable on the installed knob.
Display source:	physical switch to LCD controller source control, not a GPIO relay selector in this build. Inputs in use: HDMI (Raspberry Pi) and AV1 (composite RCA); AV2 and VGA had nothing connected at the initial build.
Composite RCA input:	video only.
Rear audio socket:	output to external speakers or headphones.
External audio:	USB capture source and audio-loop helper.
External USB access:	one rear USB-C port; the Pi's other ports are inside the case.

Software, services and recovery

The software includes Raspberry Pi OS, X11/LXDE, Openbox, LightDM, PCManFM, PipeWire, and WirePlumber. System services include apple-gesture-button, volume-knob, atease, and flyingtoasters. User services include piforma-panel, controlstrip-simulator, and the optional usb-audio-loop.

Use the canonical repository's software page, service inventory, configuration archive and installation guide for recovery. Archived helpers describe an earlier installation; do not substitute them for deployed scripts without checking the running configuration.

Keyboard conversion and sources

The build credits Matt Chesters' zmk-apple-desktop-bus-keyboard conversion. Special keys follow the deployed keymap.

The keyboard conversion uses a Seeed XIAO nRF52840, MCP23017 I/O expander, custom board, 100 mAh battery and nice!view display. Profile 2 of its five Bluetooth profiles is paired with the PiForma.

Appendix B: When Something Goes Wrong

Start with what you can see and hear. Make one check at a time, then observe the result. Note any message before dismissing it. Stop using the PiForma if there is unusual heat, swelling, a hot smell or damaged power wiring; those symptoms are a reason to speak to us, not try another restart.

No green light, chime or picture

Check the power switch on the right side. Press the rear battery plunger and read the battery percentage through the top slits. Plug in the power cable, then try again. Even a battery that showed full has sometimes needed a minute on power. If nothing works, ask us rather than opening it up.

Battery does not appear to charge

Check that the power cable is plugged in at both ends. For the bag arrangement, check the seating in the foam indents and that the cord from the bag's back pocket is plugged into a working USB charger. You can't see the battery display in the bag, so the only way to be sure the bag is charging is to unpack the computer and look. Don't swap in a different power supply.

Green light or chime, but no desktop

Allow approximately one to two minutes for startup, with some variation. A No Signal message and a moment of blank screen are normal at first. If the startup picture doesn't follow within about 15 seconds, press the rear source selector until the screen shows HDMI. The chime plays before the computer has finished starting. If the startup picture persists, note where it stops. If the image flickers or distorts, stop using the computer; do not tap the case or adjust internal cables while powered. Ask us to investigate a persistent display problem.

Mouse pointer does not move

Check the underside power switch and put the mouse on its pad. Keep the optical tracking window facing the pad. The mouse's USB wireless receiver is plugged into the Raspberry Pi inside the case, so there is nothing to check there without dismantling the computer. The Sync button is not part of normal setup. If the pointer works elsewhere but stays inside one emulator, it's probably trapped by that emulator; see Chapter 3.

Keyboard does not type

If letters come out as capitals or the number keys type symbols, Caps Lock is down: press it again so it sits level with the other keys. Otherwise, check the switch on the right-hand edge, look at the nice!view display for battery level, an X that means it isn't connected, and profile 2 selected, and click in the intended text field. Try another available text field. If typing still fails everywhere, ask us rather than trying guessed key combinations.

No sound, or no startup chime

For normal program sound, turn the side volume knob or the keyboard's volume dial up a little and check that the program has something playing and its own sound is enabled. Pressing the knob doesn't mute. Start with the volume low. If sound has been switched to the rear audio socket, it plays only through whatever is plugged in there, not the internal speaker. Switch it back with Sound Options in the Wi-Fi status menu at the far right of the menu bar.

For a missing startup chime, remember that some rear selector positions are silent. Turn the dial a click at a time with the power on, as Chapter 1 describes; not every position plays. The chime and normal program sound use separate hardware; one can work while the other doesn't. Describe which sound is missing before asking for help.

A website or network service will not open

Try another known page or service. Check the Wi-Fi status icon at the far right of the menu bar; its menu can take 20 or 30 seconds to open. If another device on that network also fails, the problem may be outside the PiForma. To try another network, click the network's name in that menu to open the wireless network manager. Note any error message, and ask us if Wi-Fi won't reconnect.

A program will not respond

Wait briefly, then look for a dialog waiting behind the main window. Check whether the pointer and other applications respond. Save unaffected work if you can and try the troubled program's own Quit or close control.

If necessary, use the six-second Apple-logo hold described in Chapter 3. Check which application is active first. Unsaved work may be lost, including game progress. Use View → Show Desktop to uncover the launcher when the menu bar is accessible.

If the entire desktop remains unresponsive, use Special → Shut Down if you can reach it. Abrupt physical power-off is a last resort and may lose work or damage files. A sleeping screen is not completed shutdown. Note what was running and what stopped responding.

If the launcher or menu bar does not return after leaving a full-screen program, save accessible work and ask us. Do not try longer logo holds.

Glossary

Apple Desktop Bus (ADB): The connection system used by the original vintage keyboard before its Bluetooth conversion.

AtEase: The PiForma's folder-like launcher for opening configured programs.

Bluetooth LE: Bluetooth Low Energy, the wireless connection used by the converted keyboard.

Chime selector: The rear dial that chooses a startup sound; some settings are silent.

Clippy: The assistant at the lower-right corner of the desktop. Type a question in its Chat with Clippy window; check its suggestions against the actual setup.

Composite video: The video signal accepted by the rear RCA input, used to show an external source on the PiForma screen. The jack carries video only, not audio.

Control Strip: The strip at the lower left of the desktop. On the PiForma it works like a dock, holding a pane for each open program and for programs kept there for quick access. It is supplied by ControlStrip Simulator.

Desktop: The working screen with its background, launchers, menu bar, and application windows.

EmulationStation: The game-list menu of RetroPie, opened from AtEase. It is a separate application from the games it starts.

Emulator: A program that imitates another computer or games system so compatible software can run.

Full-screen application: A program occupying the display, often covering the normal desktop menus.

GPIO: General-purpose input/output pins on the Raspberry Pi, used for controls in this build. Appendix A uses BCM pin numbering.

HDMI: The digital video connection that carries the Raspberry Pi's picture to the PiForma screen. Select HDMI on the rear source selector to see the Pi.

microSD: The small storage card containing the PiForma's operating system and saved files.

nice!view: The small display above the keyboard's number keys. It shows the keyboard's battery level, Bluetooth status, Bluetooth profile and typing speed. The PiForma uses profile 2.

PiForma Guide: The built-in guide to the computer, opened from AtEase. Its window is titled Macintosh PiForma Guided Tour, and its program is the Docs Viewer listed in the credits.

PiForma Panel: The program that supplies the menu bar across the top of the screen, including the Special menu with Shut Down and Sleep Display.

Qi charging: Wireless charging using an aligned transmitter and receiver.

Raspberry Pi: The small computer board inside the PiForma case.

Raspberry Pi OS: The operating system running beneath the PiForma's Macintosh-inspired desktop.

RetroPie: The games/emulation environment, opened through EmulationStation in this setup.

Save state: An emulator's snapshot of a game in progress. It is a different facility from the game's own save.

Screen saver: An animation that fills the screen. On the PiForma, the Flying Toasters are a screen saver you start yourself; nothing starts on its own when the computer is idle. Chapter 3 explains how to stop it.

Sound Options: The Wi-Fi status menu item for choosing the sound output: internal speaker or rear audio socket.

Source selector: The rear button that moves the screen to its next input. HDMI shows the Raspberry Pi; AV1 shows the composite RCA input.

USB hub: An accessory that provides several USB connections from one computer port.

Wi-Fi: The wireless network connection, chosen from the Wi-Fi status menu at the far right of the menu bar.

Credits and acknowledgments

This build uses many good ideas by other people, adapted to a computer-shaped joke that became a useful project. The Apple manuals supplied visual and organizational references, not PiForma specifications or approval. Apple had no part in the PiForma; the story in the welcome is a tall tale. Their text has not been copied into this guide.

- Our projects: github.com/milagrofrost.
- Build records: github.com/milagrofrost/Macintosh-PiForma-docs.
- This guide: github.com/milagrofrost/piforma-owners-guide.
- Keyboard conversion, by Matt Chesters: github.com/mattchesters/zmk-apple-desktop-bus-keyboard.
- Mouse shell: Retro Wireless Computer Mouse, with a custom scroll wheel and axle, both on Printables.
- PiForma Panel, the menu bar: github.com/milagrofrost/piforma-panel.
- AtEase Simulator: github.com/milagrofrost/AtEase-simulator.
- ControlStrip Simulator: github.com/milagrofrost/ControlStrip-Simulator.
- About This PiForma: github.com/milagrofrost/about-this-pi.
- PiForma Guide (Docs Viewer): github.com/milagrofrost/Macintosh-PiForma-docs-viewer.
- LimeWire Simulator: github.com/milagrofrost/limewire-simulator.
- Clippy, forked from Felix Rieseberg's project: github.com/milagrofrost/clippy.

PiForma Limited Warranty

We warrant that this PiForma was a working computer when it left the workbench. That is the whole of the warranty.

There is no repair turnaround time and no promise that anything that breaks will ever be fixed. Repairs, if any, happen when we have a free weekend and the right part. Opening the Forbidden Latch voids this warranty, which costs you nothing, because there was never much of one to void.

In keeping with the company's priorities of the era, this warranty is offered to consumers only. Enterprise customers should consult their enterprise agreement, of which there is none.

Radio and television interference

The only part of the PiForma that broadcasts anything on purpose is its Raspberry Pi, whose Wi-Fi and Bluetooth radios are certified to FCC rules. Everything else in the case should keep its radio-frequency energy to itself.

If it interferes with radio or television reception, which you can find out by switching it off and on, move it farther from the receiver. Alternatively, switch off the television and play a tape through the PiForma's rear composite input instead.

Tell us what you think

Apple's manuals ended with a reply card asking what you thought. This one does too, without the postage. If you have looked through the PiForma or this guide and have a thought, a question or a correction, tell us in person or open an issue at github.com/milagrofrost/Macintosh-PiForma-docs.

Compliments are forwarded to the CEO. Complaints are forwarded to the rogue engineer.

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